

E-Commerce Sales & Customer Analytics Dashboard

Sales Performance & Customer Behavior Analysis

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Description

This report analyzes an online retail store's sales and customer data using an interactive Power BI dashboard. It provides insights into sales distribution across product categories and regions, order channels, customer type, payment status, return status, and overall customer satisfaction.

Project Objective

The objective of this dashboard is to provide a clear overview of sales performance and customer behavior while allowing users to interactively analyze the data using category, channel, payment status, return status, and customer type filters.

Tools Used

- Microsoft Power BI
- Power Query
- DAX
- Data Visualization

Dashboard Components

- KPI cards for Total Orders, Average Order Value, and Customer Rating
- Sales by Category
- Customer Type (New vs. Returning)
- Order Channel
- Payment Status
- Return Status
- Sales by Region
- Interactive slicers for Category, Channel, Payment Status, Return Status, and Customer Type

Dashboard Analysis

1. High-Level Overview

The E-Commerce Sales & Customer Analytics Dashboard provides a single-page overview of orders, sales performance, and customer-related characteristics. The dashboard combines KPI cards, charts, and interactive slicers to help users understand overall sales trends and identify important patterns across categories, channels, and regions.

2. KPI Analysis

KPI	Value	Interpretation
Total Orders	12,480	The dashboard contains a total of 12,480 orders across the available e-commerce dataset.
Average Order Value	₹2,340	The overall average order value is ₹2,340, providing a quick indication of typical customer spending.
Customer Rating	4.3 / 5	The overall average customer rating is 4.3 out of 5, indicating generally strong customer satisfaction.

3. Chart Analysis

Sales by Category

The Sales by Category chart compares order volume across different product categories. It helps identify categories with larger and smaller sales contributions and provides an overview of how sales

are distributed across the catalog.

Customer Type

The Customer Type chart shows the split between new and returning customers. This provides insight into how much of the sales volume is driven by repeat business versus new customer acquisition.

Order Channel

The Order Channel chart compares orders placed online versus in-store. This helps understand which channel drives the majority of transactions.

Payment Status

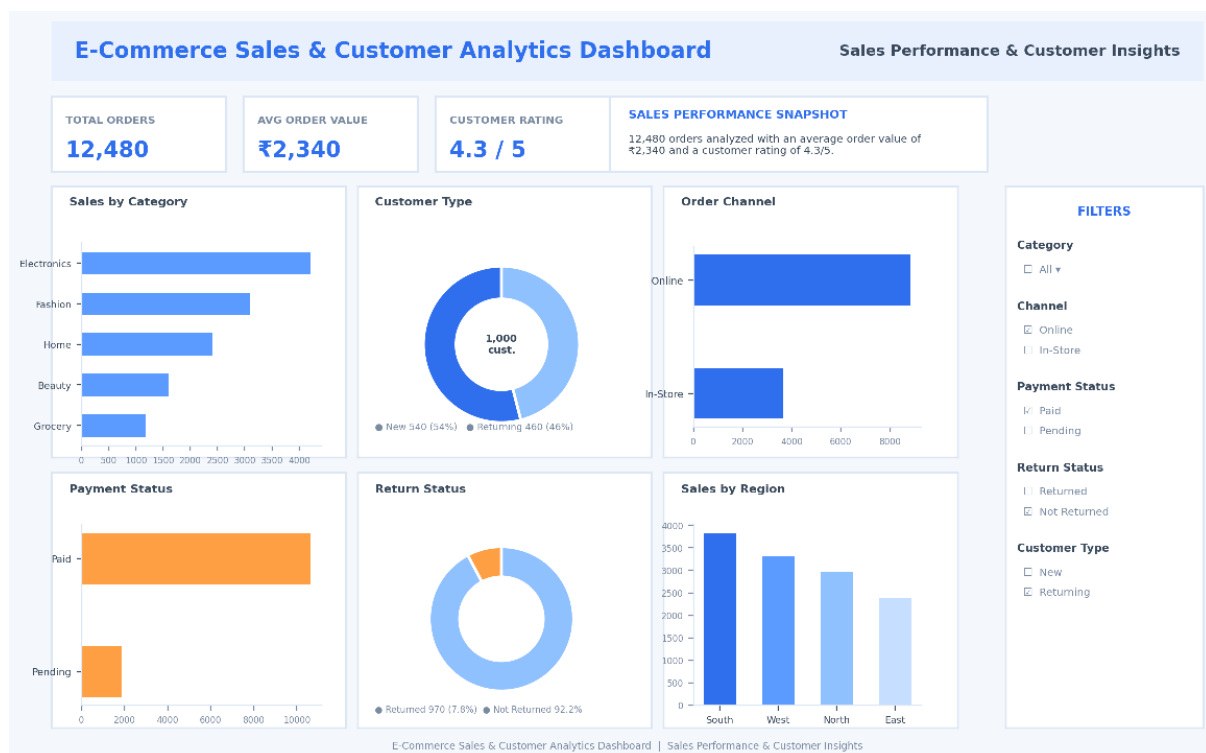
The Payment Status chart compares orders based on payment completion, helping identify the number of orders that are paid versus pending.

Return Status

The Return Status chart presents the proportion of orders that were returned versus not returned. It provides a quick overview of product and fulfillment quality.

Sales by Region

The Sales by Region chart shows order volume across the four sales regions. This can help identify high-performing and underperforming regions for targeted strategy.



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Insights, Recommendations & Summary

4. Key Insights

Sales Performance: A total of 12,480 orders were recorded with an average order value of ₹2,340, indicating healthy overall transaction volume.

Customer Satisfaction: The overall average customer rating is 4.3 out of 5, suggesting that customers are generally satisfied with their purchases.

Category Distribution: Order volume varies across product categories, with Electronics and Fashion among the top-performing categories in the dataset.

Channel Preference: Online orders significantly outnumber in-store orders, highlighting the dominance of the digital sales channel.

Customer Loyalty: The dashboard shows a fairly balanced mix of new and returning customers, with a slight lean toward new customer acquisition.

Fulfillment Quality: Returns account for a small proportion of total orders, indicating relatively strong product and delivery quality.

5. Recommendations

- Identify underperforming product categories and evaluate pricing or promotional strategies.
- Monitor pending payments and set up automated reminders to reduce payment delays.
- Analyze return status alongside category and channel data to identify root causes of returns.
- Invest in retention campaigns to convert new customers into returning customers.
- Compare regional sales performance to guide inventory and marketing allocation.
- Use the interactive slicers to identify specific customer or order segments requiring attention.
- Track sales performance and customer ratings over multiple periods if historical data becomes available.

6. Closing Summary

Overall, the E-Commerce Sales & Customer Analytics Dashboard provides a comprehensive view of sales and customer behavior by combining order, channel, payment, return, and regional information. The dashboard makes it easier for business stakeholders to identify patterns, compare categories and regions, and understand customer engagement. The interactive slicers further allow users to investigate specific segments and make data-driven decisions.

7. Data Notes

Source: E-commerce sales dataset containing order, category, channel, payment, return, and regional information.

Tools: Microsoft Power BI, Power Query and DAX.

Note: All values and charts presented in this report are generated from the sales dataset used to build the Power BI dashboard.